

Categories, Proofs, and Processes:

Chapter 3 Limits

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3 Limits and Colimits

3.1 Equalisers

3.1.1 Motivation

In **Set**, we can define equality between functions f, g when $\forall x \in X, f(x) = g(x)$, but we might be interested in situations where where $f(x) = g(x)$ for $x \in Y \subsetneq X$.

For example, $X = \{0, 1\}^*$, but only a subset G could encode graphs using this alphabet.

3.1.2 Definition

Definition 3.1. Consider a pair of parallel arrows $A \begin{smallmatrix} f \\ \rightrightarrows \\ g \end{smallmatrix} B$. An **equaliser** of (f, g) is an arrow $e : E \rightarrow A$ such that

$$f \circ e = g \circ e$$

and for any $h : D \rightarrow A$ such that $f \circ h = g \circ h$, there is a unique $\hat{h} : D \rightarrow E$ so that

$$h = e \circ \hat{h}$$

Diagrammatically,

$$\begin{array}{ccc} E & \xrightarrow{e} & A \\ \uparrow \hat{h} & \nearrow h & \uparrow \\ D & & A \end{array} \quad \begin{array}{c} f \\ \rightrightarrows \\ g \end{array} B$$

Note 3.2. A diagram commutes if any two paths with a common source and target, with at least one of the paths at least length > 1 , are equal.

Example 3.3. In **Set**, the equaliser of f, g is given by the inclusion

$$\{x \in A \mid f(x) = g(x)\} \hookrightarrow A$$

For $f, g : \mathbb{R} \rightarrow \mathbb{R}$ where

$$\forall x, f(x) = ax^2 + bx + c, g(x) = 0$$

the equaliser is the roots of $f(x)$ if they exist.

3.1.3 Results

Proposition 3.4. *Any equaliser in any category is monic.*

Proof.

$$\begin{array}{ccccc} E & \xrightarrow{e} & A & \xrightarrow[f]{g} & B \\ x \uparrow \parallel y & \nearrow h & & & \\ D & & & & \end{array}$$

Suppose $e \circ x = e \circ y$, then it is equal to h . Then

$$f \circ h = f \circ e \circ x = g \circ e \circ x = g \circ h$$

There is a unique map $h' : D \rightarrow E$ such that $e \circ h' = h$. But since

$$e \circ x = h \wedge e \circ y = h \implies x = y = h'$$

by uniqueness, so e is monic. □

Definition 3.5. **Coequaliser** in $\mathcal{C}$ is an equaliser in $\mathcal{C}^{op}$.

Proposition 3.6. *Any coequaliser in any category is epic.*

Proposition 3.7. *Equalisers and coequalisers are unique up to a unique isomorphism.*

Proof. Suppose

$$\begin{array}{ccc} E_1 & \xrightarrow{e_1} & A \xrightarrow[f]{g} B \\ & \nearrow e_2 & \\ E_2 & & \end{array}$$

Thus there are unique maps

$$\begin{aligned} u_1 : E_2 &\rightarrow E_1, & e_2 &= e_1 \circ u_1 \\ u_2 : E_1 &\rightarrow E_2, & e_1 &= e_2 \circ u_2 \end{aligned}$$

Then identities exist

$$\begin{aligned} 1_{E_1} : E_1 &\rightarrow E_1 := u_1 \circ u_2 \\ 1_{E_2} : E_2 &\rightarrow E_2 := u_2 \circ u_1 \end{aligned}$$

□

3.2 Pullbacks

3.2.1 Definitions

Definition 3.8. Given a pair of morphisms

$$A \xrightarrow{f} C \xleftarrow{g} B$$

(called a cospan), we define an (f, g) -cone to be

$$A \xleftarrow{p} D \xrightarrow{q} B$$

such that the following diagram commutes.

$$\begin{array}{ccc} D & \xrightarrow{q} & B \\ p \downarrow & & \downarrow g \\ A & \xrightarrow{f} & C \end{array}$$

Definition 3.9. The ***pullback*** of f along g is an (f, g) -cone such that for any other (f, g) -cone $A' \xleftarrow{p'} D' \xrightarrow{q'} B$, there is a unique $h : D' \rightarrow D$ such that $p' = p \circ h, q' = q \circ h$. Diagrammatically,

$$\begin{array}{ccccc} D' & & & & \\ & \searrow^{q'} & & & \\ & & D & \xrightarrow{q} & B \\ & \searrow^{p'} & \downarrow p & & \downarrow g \\ & & A & \xrightarrow{f} & C \end{array}$$

(Note: A dashed arrow h points from D' to D in the original diagram.)

Note 3.10. Pullback often denoted as

$$A \times_C B$$

Example 3.11. In **Set**, pullback is subset of cartesian product. Take a function $f : A \rightarrow B$ and subset $V \subseteq B$. Consider

$$f^{-1}(V) = \{a \in A \mid f(a) \in V\} \subseteq A$$

Consider

$$\begin{array}{ccc} f^{-1}(V) & \xrightarrow{\bar{f}} & V \\ \downarrow & & \downarrow \\ A & \xrightarrow{f} & B \end{array}$$

So $f^{-1}(V)$ is a pullback, unique up to a unique isomorphism.

Proposition 3.12. *Pullbacks are unique up to a unique isomorphism.*

3.2.2 Results

Proposition 3.13. *If a category $\mathcal{C}$ has binary products and equalisers, then it has pullbacks.*

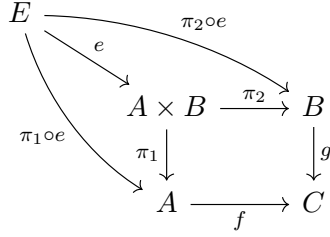
Proof. Given the cospan

$$A \xleftarrow{p} C \xrightarrow{q} B$$

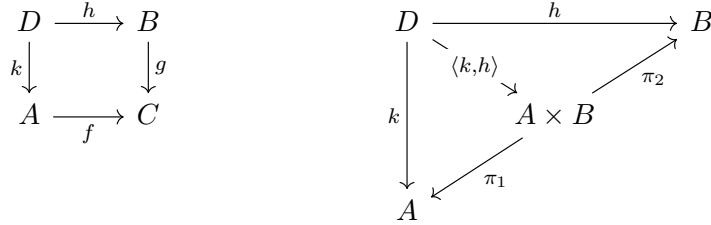
we want to construct its pullback using products and equalisers. Consider the product

$$A \xleftarrow{\pi_1} A \times B \xrightarrow{\pi_2} B$$

Form an equaliser $e : E \rightarrow A \times B$. The following illustrative diagram does not commute



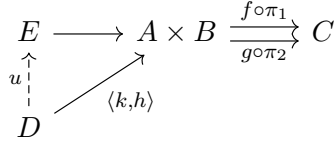
We want to show $A \xleftarrow{\pi_1 \circ e} E \xrightarrow{\pi_2 \circ e} B$ is a pullback, since e might not be unique. Assume the following diagrams commute



Then

$$f \circ \pi_1 \circ \langle k, h \rangle = g \circ \pi_2 \circ \langle k, h \rangle$$

Combine this with the original equaliser and we have



Unique map u , thus $A \xleftarrow{\pi_1 \circ e} E \xrightarrow{\pi_2 \circ e} B$ is a pullback. □

3.3 Summary

Constructs follow similar pattern

1. Determine shape of a diagram
2. Highlight an object and arrows in that diagram
3. Stipulate unique map to (from) that object from (to) all other objects with the same "composition" properties

Note 3.14. We are exhibiting a universal (mapping) property where the unique map is a "unique morphism".

3.4 Limits